

Difference constraints

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Description:

Given a set of $a_i - b_j \leq c_k$, determine does it exist a set of a_i and b_i to fulfill the constraints

Algorithm

The idea is come from shortest path,if you regard a_i as d_v ,and b_i as d_u and c_k as w ,then you will get $d_v \leq d_u + w$,in a graph given source s ,at the end for every vertex (u, v) and $u \rightarrow v$ we have $d_u + w \geq d_v$,so this is completely same as the problem above,so given $a_i - b_j \leq c_k$ we can just make an edge $j \rightarrow i$ with weight c_k ,since the weight may be negative so we need spfa/bellman ford,if exist negative cycle,it can be show that no such solution(contradiction),at the end,set of d_i is one of the set of solution to make the constraints establish,and also it can be extend,if we need to restrict $L \leq a_i \leq R$ then we can use a special node z ,and define $a_z + L \leq a_i \leq R + a_z$ and get two constraints, $a_z - a_i \leq -L$ and $a_i - a_z \leq R$,and after than let $d_z \neq 0$ then all vertex i including z to do $d_i + = d_z$,then it can be show that this is one of the solution

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